FRACTIONAL RATE DECIMATOR AND INTERPOLATOR DESIGN

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1 ABSTRACT

In this article fractional rate sampling frequency change is considered. A general theory based on combined analog and digital interpolators is presented, a generalized Farrow structure is introduced, and comparisons between cases with different design parameters are given. It is shown that the introduction of the digital interpolator provides simplifications in terms of computational complexity at the cost of increased coefficient storage.

2 INTRODUCTION

Changing the sampling frequency becomes necessary whenever two systems running at different sampling frequencies are to be interconnected. Another reason for changing the sampling rate is to keep the rate as low as possible for any algorithm in a system to minimize the overall computational complexity. In this case it is imperative that the the sampling rate change operations are less complex than what is saved by the rate change.

The term *fractional* indicates that the ratio between the sampling frequencies to be interfaced is a fractional number. Two cases that have fundamentally different solutions can be distinguished. The simplest case is when the rate ratio is given by a rational number. Then an integer factor interpolator is applied, and the interpolated signal is decimated by calculating only the wanted output samples. It becomes somewhat more involved when the rate ratio is an irrational number, and this is the main topic of the paper. The rational case is a special case of the irrational case, and any method developed for the general case can therefore always be used for the rational case.

For fractional rate change, the distinction between interpolation and decimation is no longer important. The only reason for keeping both is to distinguish between an increase or a decrease of the sampling frequency. In all cases we have to find samples between the original samples, as in integer factor interpolation. In the sequel the term *fractional interpolation* is used in all cases.

The main technique in fractional interpolation is the use of time-varying filters to produce samples at appropriate times. For the two classes mentioned above the following techniques can be applied:

1. *Rational ratios*: Store a number of filter coefficient sets and apply the correct set based on timing.
2. *Irrational ratios*: Calculate new coefficients as required for each new sample.

In the next section we give a unified description of the different techniques based on analog filter reconstruction and resampling plus digital interpolation combined with the analog technique.

Fractional rate interpolation has many applications. The first work in this field was done to interface audio systems with different sampling frequencies [2, 3, 4]. In digital communication systems this is also important, although then equalization of phase differences is often the central problem. In echo cancelation based on subband decomposition some oversampling is necessary to avoid aliasing when manipulating the amplification of the different bands before recombining them. Also it is possible to construct very general sampling rate changers based on the type of techniques presented here.

In this paper only FIR-based systems are considered. IIR filters can also be used, see [3].

3 THEORY BASED ON ANALOG RECONSTRUCTION AND RESAMPLING

Any digital signal can be interpolated by first reconstructing it to analog form, followed by resampling at any new rate. We shall use this as our theoretical basis, and in fact mimic this procedure for the proposed techniques applicable to irrational ratio interpolation.

Given a digital signal $x(n)$, reconstruction can be obtained by applying an analog reconstruction filter with impulse response $f(t)$, as

$$y(t) = \sum_{k=-\infty}^{\infty} f(t - kT)x(k), \tag{1}$$

where T is the sampling interval of the input signal.

It may preferable to apply a digital interpolation filter prior to the analog reconstruction to make the analog

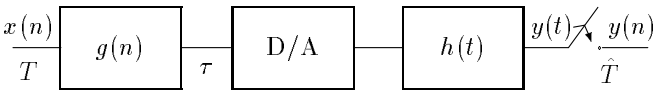


Figure 1: Interpolation system including digital interpolator

filter simpler [3], as shown in Figure 1. The digital interpolation filter is designed for N times oversampling. Its unit sample response is $g(n)$. The sampling interval after digital interpolation is denoted $\tau = T/N$. When combining the digital and analog filters, an equivalent filter will have the impulse response

$$f(t) = \sum_n h(t - n\tau)g(n). \tag{2}$$

The output signal is computed by Equation 1 keeping in mind that $T = N\tau$.

To compute a sample at an arbitrary instant requires that either all necessary values of the impulse response are stored or that we calculate the required samples of $f(t)$. Exact storing is obviously impossible because an infinite number of samples will be required. However, if N is selected to be extremely large, the resampling will simply mean to pick the sample closest to the wanted time instant [2]. A worst-case analysis of this method for audio applications indicates that an oversampling factor of close to 10^5 is necessary [3]!

In general, we want to find any output sample, whether it is resampling at a new rate, phase shift, or some random sampling. Irrespective of the application, it is practical to express the sampling time t_i in relation to the digitally interpolated samples at $p\tau$ as

$$t_i = (p + \epsilon)\tau, \ \ \epsilon \in [0, 1). \tag{3}$$

p can be found as $p = \lfloor t_i/\tau \rfloor$.

Let us calculate the unit sample response of the composite filter expressed in terms of p and ϵ.

$$\begin{aligned} f\big((p+\epsilon)\tau\big) &= \sum_n h\big((p-n+\epsilon)\tau\big)g(n) \\ &= \sum_{l\in\mathcal{K}} h\big((l+\epsilon)\tau\big)g(p-l) \end{aligned} \tag{4}$$

In the last expression the summation limits are given by the set $\mathcal{K}$. This set is found from the length of the analog impulse response $h(t)$ given by $J\tau$, and the length of the digital filter, K. The composite impulse response has length $(K + J - 1)\tau$ and the set over which the sum is taken is $\mathcal{K} = \{\max(0, p - (K - 1)), \min(p, J - 1)\}$.

Let us revisit the rational rate interpolation case before we study practical solutions for irrational ratios based on the above theory.

Assume that the output sampling interval is $\hat{T} = (M/N)T$. It is then clear that $l\hat{T} = l(M/N)T$ can generate N different phases between the input samples, implying that N coefficient sets will suffice to find any of the required samples.

4 METHODS BASED ON POLYNOMIAL INTERPOLATION

As derived in the previous section, exact fractional interpolation by irrational factors requires recalculation of the filter coefficients for every new output sample. It is therefore imperative that the coefficient calculations be simple. One measure for simplifying the these calculations is to use polynomial functions to describe the unit sample response of the analog reconstruction filter. By doing so, only multiplications and additions are necessary to find the coefficients. Such functions include Lagrange interpolators [3], spline functions, or more general polynomial series [6].

A good interpolator should have an impulse response that for sample values at $t = k\tau$ is equal to 1 for one sample and equal to 0 otherwise. This property guarantees that the original samples are not changed during the interpolation. The detailed shape of the impulse response between these samples is, of course, also impotant for the filter performance.

4.1 Taylor series expansion

A very general way of designing a polynomial based analog impulse response is to expand each sample interval τ in a Taylor series. The more terms in the series, the better is the approximation to any shape of the impulse response.

Assume that we write the impulse response as

$$h\big((l+\epsilon)\tau\big) = \sum_{r=0}^{R} c_r(l)\epsilon^r, \ \ \epsilon \in [0, 1), \ \ l \in [0, J-1], \tag{5}$$

where for convenience we have set $\tau = 1$ on the right-hand side of the equation. Notice that there are different series expansions for each interval τ.

The impulse response of the composite filter now becomes

$$\begin{aligned} f\big((p+\epsilon)\tau\big) &= \sum_{l\in\mathcal{K}} \sum_{r=0}^{R} c_r(l)\epsilon^r g(p-l) \\ &= \sum_{r=0}^{R} \epsilon^r \sum_{l\in\mathcal{K}} c_r(l)g(p-l) \\ &= \sum_{r=0}^{R} \epsilon^r w_r(p), \end{aligned} \tag{6}$$

which shows that it is a series expansion in ϵ. The "coefficients" in this expansion are functions of p and are constructed as a convolution between the unit sample response of the digital filter and the expansion coefficients for the different intervals of the impulse response of the analog filter.

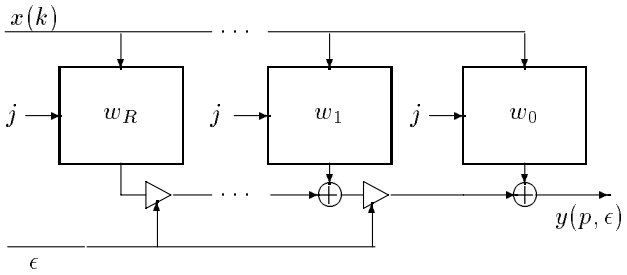


Figure 2: Time-varying Farrow-structure

Excellent frequency responses have been obtained by optimizing the parameters given by the above theory [6].

5 STRUCTURES FOR FRACTIONAL RATE INTERPOLATORS

Two equivalent, but different structures for implementing the fractional interpolation system are suggested.

The first structure is a direct replica of Figure 1, as presented in [3, 4]. The analog filter coefficients are calculated from a formula, which in [3] was a Lagrange interpolator, but could also be any function of the form in Equation 5. The efficiency of this structure relies on the fact that when the analog filter is of length $J\tau$, only J consequtive samples from the digital interpolator is required to produce one output sample. The remaining samples are never calculated. The experience from using this structure is that a first-order Lagrange interpolator is often a very good choice. The required performance is obtained by selecting the oversampling factor and optimizing the coefficients of the digital interpolator.

The second structure is a generalization of the Farrow structure [1] derived from the composite filter. If we go back to the Equation 1, and notice that the input sampling interval is given by $T = N\tau$, we see that the output signal is calculated using one Nth of the filter coefficients of the composite filter. Inserting $t = (p+\epsilon)\tau$ in Equation 1, and furthermore applying the impulse response in Equation 6, we obtain

$$\begin{aligned} y\big((p+\epsilon)\tau\big) &= \sum_k f\big((p+\epsilon-kN)\tau\big)x(k) \\ &= \sum_{r=0}^{R} \epsilon^r \sum_{k\in\mathcal{K}_\mathcal{N}} w_r(p-kN)x(k). \end{aligned} \tag{7}$$

This equation tells us that the we can organize the filter coefficients for each r in N distinct sets: $w_r(j,k) = w_r(j-kN)$, $j = p_{\bmod N}$, $k \in \mathcal{K}$.

The resulting generalization of the Farrow structure is shown in Figure 2.

The choice between N coefficient sets controlled by j makes the filter a time-varying system.

6 FILTER EXAMPLES

The above theory offers abundant flexibility in the choice of parameters. In this section we illustrate the proposed method and especially demonstrate the potential of oversampling.

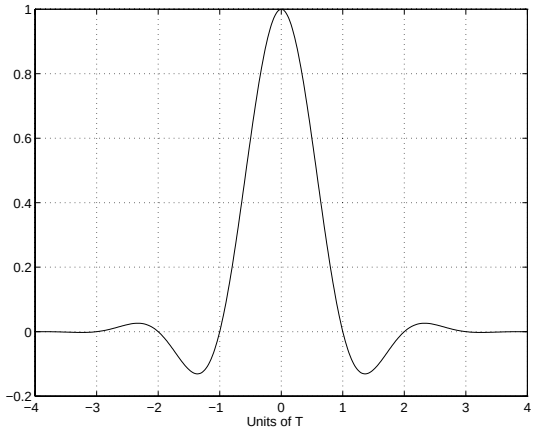


Figure 3: Impulse response of analog interpolation model filter

In the first example we illustrate how well we can approximate an analog interpolation filter by a composite response. The desired analog reconstruction filter $f(t)$ is given as a sinc-function multiplied by a Blackman window, as shown in Figure 3. The approximation is obtained by sampling this function regularly and making a continuous approximation to the desired response by adding linear segments between the samples.

The fidelity of the approximation is illustrated in Figure 4 as the difference between the exact response and the linear segment approximation for 5 and 10 times oversampling.

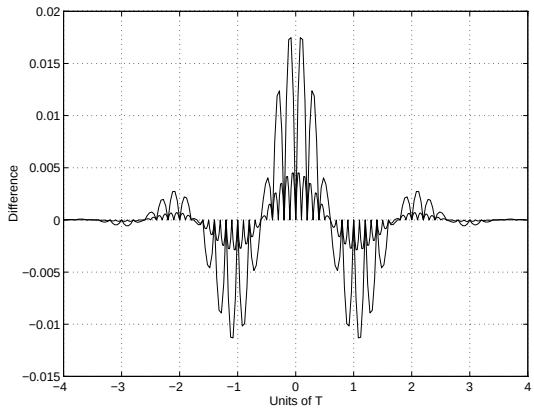


Figure 4: Difference between the exact analog model impulse response and the response composed of linear segments between samples. The function with the larger values represent the 5 times oversampling case while the other represents the 10 times oversampling case.

How can the segmented response be implemented in the two suggested systems? For the system in Figure 1, the approximate response can be obtained by cascading

a digital filter with unit sample response obtained as above and an analog linear interpolator, that is a filter with a symmetric and triangular response. To calculate an output sample in the linear interpolator, one sample on either side of the sampling instant must be provided by the digital interpolator. With N times oversampling, the digital interpolator requires $2Kf_s/N$ multiplications per second when $f_s = 1/T$ is the input sampling frequency. In addition, one multiplication is used to calculate the weighted average of the two samples in the "analog" filter.

Because the Taylor series is of first degree, the Farrow structure only has two subfilters. The two subfilters have approximately K/N multipliers each, and one extra multiplication by ϵ is needed to obtain the output signal. Altogether, the complexity in terms of multiplications is approximately equal for the two systems considered.

The difference functions would be improved significantly by adding second degree polynomials of the form $c\epsilon(1-\epsilon)$ in each interval, or at least for the intervals with the largest error. This would alleviate the need for high oversampling rates, but would add another subfilter to the Farrow structure. This means a reduction in the coefficient storage requirement, but approximately 50% increase in computations.

As a second example we maintain 10 times oversampling in the digital filter and the linear interpolation for the analog filter, but this time we optimize the digital interpolator using the Remez algorithm. The filter is specified to have a transition band in $[0.35, 0.65]f_s$, and a stopband attenuation better than 50 dB.

The frequency response of the composite filter is shown in Figure 5.

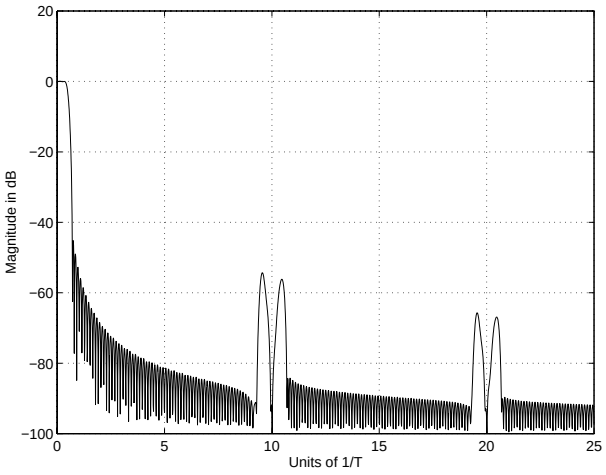


Figure 5: Frequency response of analog filter composed of digital filter and analog linear interpolator

The system in [5] does not use oversampling, but has similar performance. The proposed structure requires approximately half the number of operations at the cost of an increase in the coefficient memory by a factor of 5. The memory increase is due to the 10 times oversampling and half the number of sub-filters. Further simplifications can be obtained by using a different number of subfilters for the different parts of the impulse response. The tradeoff between the order of the analog filter and the oversampling must be evaluated for each application.

7 CONCLUSION

The structure suggested in this paper is a generalization of the Farrow structure [1], and is an alternative to the two-stage structure suggested in [3]. Oversampling in the Farrow structure simplifies the computations, but there is basically no advantage over the two-stage structure. The choice of analog interpolation function is obviously important. The Lagrange interpolation function chosen in [3] leaves little room for optimization. However, in the example presented in this paper a first degree polynomial is used, and this coincides with Lagrange interpolation. With higher order analog filters it is better to use different polynomial approximations for each sampling interval. And unlike Lagrange interpolators, it is then possible to use second degree polynomials.

The Farrow structure lends itself to hardware implementation due to its regularity. As such it is better than the two-stage implementation.

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